Design and Analysis of a Low-Power Conditional Capture Flip-Flop in 32 nm CMOS

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***Abstract*—This paper presents the design and analysis of a low-power conditional capture flip-flop (CCFF) implemented in a 32 nm CMOS technology. The proposed CCFF reduces redundant internal switching when the input data remains unchanged, thereby achieving significant clock power reduction while maintaining competitive timing performance. Both pre-layout and post-layout simulations are performed to evaluate power consumption, timing characteristics, and layout area. The obtained results demonstrate the effectiveness of the proposed design and are compared with previously published low-power flip-flop architectures.**

***Index Terms*—Conditional capture flip-flop, low power, clock power reduction, 32 nm CMOS, timing, layout, parasitic extraction.**

I. Introduction

FLIP-FLOPS are fundamental building blocks in synchronous digital systems and are extensively used in registers, pipelines, and memory elements. With the continuous scaling of CMOS technology, modern integrated circuits employ an increasingly large number of flip-flops to achieve higher performance through deep pipelining. As a result, power consumption associated with sequential elements and the clock distribution network has become a major concern. Several studies have shown that clocking-related power can account for approximately 30–50% of the total dynamic power in contemporary VLSI systems.

Conventional master–slave and pulse-triggered flip-flops suffer from significant unnecessary internal switching activity, even when the input data remains unchanged across consecutive clock cycles. This redundant switching leads to excessive clock power dissipation without contributing to useful computation, especially in low data activity scenarios where the probability of data transitions is relatively small. Therefore, reducing clock power while maintaining acceptable timing performance has become an important design objective in low-power digital circuits.

To address this issue, several low-power flip-flop techniques have been proposed, including conditional precharge, conditional discharge, and conditional capture mechanisms. Among these approaches, Conditional Capture Flip-Flops (CCFFs) exploit data dependency by preventing internal switching activity when the input data is equal to the stored output value (D = Q). By selectively disabling unnecessary transitions, CCFFs achieve statistical power reduction while preserving correct functionality and competitive timing characteristics.

In this project, a low-power single-ended Conditional Capture Flip-Flop is designed and analyzed using a 32 nm CMOS technology. The proposed design employs simplified control logic to minimize redundant internal switching and reduce clock power consumption. The flip-flop is implemented at the transistor level using Electric for schematic and layout design. Comprehensive pre-layout and post-layout simulations are performed to evaluate timing parameters, power consumption, and area overhead. Furthermore, the proposed CCFF is compared against a conventional flip-flop and previously published low-power flip-flop designs to demonstrate its effectiveness in achieving clock power reduction with reasonable performance trade-offs.

II. Related Work

Flip-flops play a critical role in determining the overall power consumption and performance of synchronous digital systems. As a result, numerous low-power flip-flop designs have been proposed over the years to reduce clock-related power dissipation while preserving reliable timing characteristics. Conventional master–slave flip-flops, although robust, suffer from unnecessary internal switching activity since internal nodes toggle on every clock edge regardless of data activity. This behavior leads to excessive dynamic power consumption, particularly in low data switching environments.

To address this issue, several data-dependent flip-flop techniques have been introduced. One of the earliest and most influential approaches is the Conditional Capture Flip-Flop (CCFF), which aims to suppress redundant internal transitions when the input data remains unchanged. Kang *et al.* demonstrated that by preventing internal switching when $D = Q$, significant statistical power reduction can be achieved without altering the functional behavior of the flip-flop. This approach is particularly effective in applications where data activity is low or highly correlated.

Another related technique is the Conditional Discharge Flip-Flop (CDFF), proposed by Mahmood *et al.*, which reduces power consumption by disabling unnecessary discharge paths during clock transitions. While CDFF designs offer improved performance and power efficiency, they often rely on more complex control logic and additional transistors, which can increase design complexity and layout overhead. Pulse-triggered flip-flop designs with conditional discharge mechanisms have also been reported, providing high-speed operation with reduced power; however, these designs may suffer from increased sensitivity to process variations and tighter timing margins.

Compared to these approaches, the CCFF technique offers a balanced trade-off between power reduction, timing robustness, and implementation complexity. In this work, a simplified single-ended CCFF architecture is adopted to further reduce transistor count and layout complexity while

maintaining the fundamental benefits of conditional capture. The proposed design builds upon the principles introduced in prior CCFF and CDFF structures, with a focus on practical implementation using transistor-level design, layout, and post-layout evaluation in a 32 nm CMOS technology.

III. Proposed CCFF Architecture

A. Conventional Master–Slave Flip-Flop (Baseline)

As a reference for evaluating the proposed low-power design, a conventional master–slave flip-flop is implemented and used as a baseline. The baseline flip-flop follows a standard static CMOS architecture and is designed using the same 32 nm CMOS technology, supply voltage, and simulation conditions as the proposed CCFF to ensure a fair comparison. In this structure, the master latch samples the input data during the active clock phase, while the slave latch holds the data and updates the output on the opposite clock phase.

Fig. 1 shows the transistor-level schematic of the baseline flip-flop. Due to its conventional operation, internal nodes within both the master and slave stages experience switching activity on every clock transition, regardless of whether the input data changes. This behavior results in redundant internal transitions and increased clock-related dynamic power consumption, particularly when the input data activity is low.

Pre-layout simulations are performed to characterize the timing and power performance of the baseline flip-flop. The measured clock-to-Q delay exhibits a rise time of approximately 3.39 ns and a fall time of approximately 23.45 ns. The output transition times are measured to be around 3.45 ns for the rising edge and 2.04 ns for the falling edge. These results reflect the timing behavior of a robust but power-inefficient conventional flip-flop structure.

Power analysis is conducted by monitoring the supply current during switching activity. The average core current is measured to be approximately 31.2 μA, resulting in an average power consumption of about 28.0 μW. The total energy consumed per operation is calculated by integrating the product of supply voltage and current over the evaluation window, yielding an energy consumption of approximately 3.37 pJ. These values serve as a baseline reference for quantifying clock power reduction and energy savings achieved by the proposed CCFF design.

B. Proposed Single-Ended CCFF

The proposed design is a single-ended conditional capture flip-flop (CCFF) that reduces unnecessary internal switching activity by exploiting data dependency. The flip-flop prevents redundant transitions when the input data remains unchanged (D = Q), thereby lowering clock-related power consumption while maintaining correct data storage and acceptable timing behavior.

Fig. 3 shows the initial CCFF schematic before optimization. Based on transistor-level analysis and extensive simulations, several targeted design refinements are introduced to improve speed, stability, and robustness. The final optimized CCFF schematic is presented in Fig. 4.

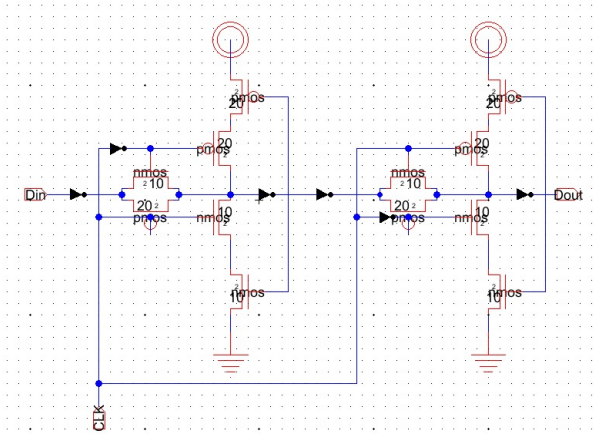


Fig. 1. Baseline master-slave flip-flop schematic.

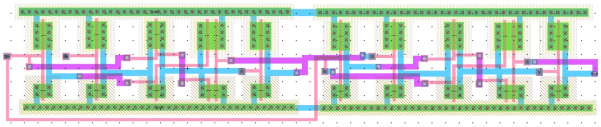


Fig. 2. Baseline flip-flop layout.

To enhance the performance of the critical data propagation path from the input D to the output Q, the transistors along the main data write path are strengthened. These devices directly affect the charging and discharging speed of internal nodes that determine the clock-to-Q delay. Increasing their effective drive strength reduces the on-resistance of the data path, enabling faster transitions and improving both clock-to-Q delay and setup time.

In addition, the transistors associated with the feedback and conditional evaluation path are carefully resized to reinforce state retention after data capture. Strengthening this path improves noise immunity and suppresses potential glitches following transitions. The sizing is optimized to maintain a balance between reliable state holding and write ability, ensuring that the feedback path does not dominate the data path during switching events.

Furthermore, a weak keeper is introduced at an internal dynamic node to prevent floating behavior during inactive evaluation phases. This node is susceptible to leakage and noise due to its dynamic nature. The keeper helps retain the stored charge and stabilizes the node when switching activity is suppressed. It is intentionally designed to be weak so that it improves robustness without interfering with normal data transitions.

Overall, the proposed sizing strategy and keeper insertion result in a well-balanced CCFF design that improves stability and timing performance with minimal power overhead. These architectural refinements form the basis for the power and performance improvements demonstrated in the subsequent simulation results.

IV. Pre-Layout Simulation Results

Pre-layout simulations are performed at the schematic level to verify the functionality and evaluate the timing and power

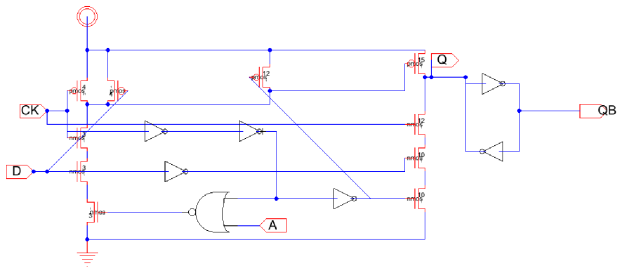


Fig. 3. Initial CCFF schematic.

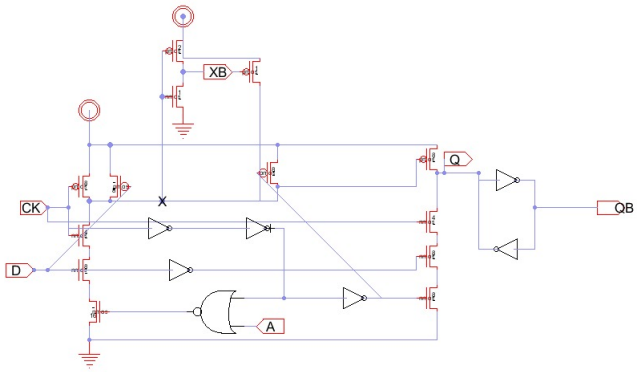


Fig. 4. Optimized CCFF schematic.

performance of the proposed flip-flop design. All simulations are carried out using a 32 nm CMOS technology with a supply voltage of $V_{DD} = 0.9$ V. Transient analysis is used to observe the dynamic behavior of the circuit under clocked operation.

Fig. 5 shows representative transient waveforms of the clock signal, input data, and output node. The results confirm correct edge-triggered operation, where the output updates only on the active clock edge and remains stable when the input data does not change.

Timing parameters are extracted from the transient waveforms using the 50% V_{DD} reference points. The measured clock-to-Q delay, output rise time, and output fall time demonstrate acceptable timing performance for the proposed design. Both rising and falling transitions are evaluated to ensure balanced operation.

Power performance is analyzed by monitoring the supply current during switching activity. The average supply current is used to compute the average power consumption, while the total energy is obtained by integrating the product of supply voltage and current over the evaluation window. These results provide a baseline for comparison with optimized designs and further analysis.

V. Layout Implementation

The physical layout of the proposed conditional capture flip-flop (CCFF) was designed and implemented using the Electric VLSI design system in a 32 nm CMOS technology. The layout process was carried out at the transistor level, following standard full-custom design practices to ensure correctness, compact area, and compatibility with post-layout extraction and simulation.

TABLE I
Pre-layout schematic simulation results ($V_{DD} = 0.9$ V).

Parameter	Value
Clock-to-Q delay (rise)	3.39 ns
Clock-to-Q delay (fall)	23.45 ns
Output rise time	3.45 ns
Output fall time	2.04 ns
Average supply current	31.2 μA
Average power consumption	28.0 μW
Total energy per operation	3.37 pJ

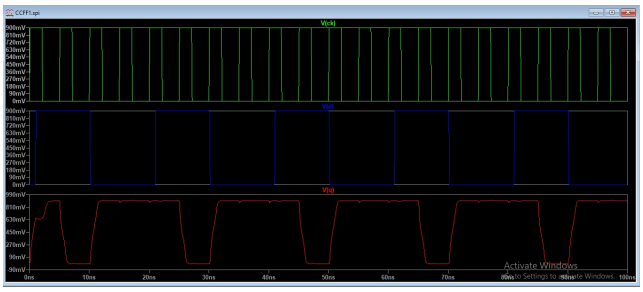


Fig. 5. Pre-layout transient waveform.

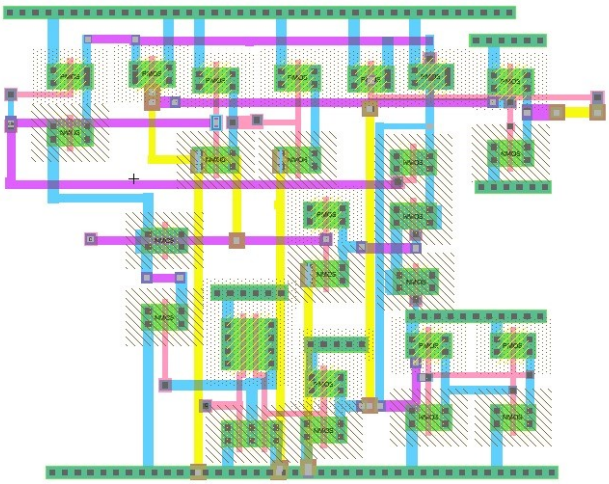


Fig. 6. Layout of the proposed CCFF in 32 nm CMOS technology.

A. Layout Methodology

The layout was developed starting from the finalized schematic, where device sizing and connectivity were carefully preserved. PMOS transistors were placed in the upper region of the layout close to the power supply rail, while NMOS transistors were placed near the ground rail to minimize diffusion breaks and simplify routing. Transistors belonging to the same functional blocks were grouped together to reduce interconnect length and parasitic effects.

To ensure correct functionality, signal routing was organized to minimize critical path length between the data input, internal storage nodes, and the output node. Special attention was given to internal dynamic nodes, where routing was kept short and well-shielded to improve robustness against leakage and noise.

B. Metal Routing and Interconnect Strategy

Multiple metal layers were utilized to achieve efficient routing and reduce congestion. Metal 1 was primarily used for local connections and short intra-cell routing between transistors. Metal 2 was employed for longer horizontal interconnects, including clock and data distribution paths, while Metal 3 was used selectively for vertical routing and power distribution to improve routing flexibility.

Contacts and vias were inserted appropriately to ensure reliable connections between diffusion regions, polysilicon gates, and metal layers. Via placement was carefully managed to avoid excessive parasitic resistance and to maintain design rule compliance across all layers.

C. Area Considerations

The layout of the proposed conditional capture flip-flop was optimized to achieve a compact footprint while preserving correct functionality and acceptable timing performance. Compact transistor placement, diffusion sharing, and efficient routing were employed to reduce layout overhead introduced by the conditional capture logic.

The optimized CCFF occupies an active layout area of approximately $44{,}880\ \mu\text{m}^2$, which represents a significant reduction compared to the conventional master–slave flip-flop baseline area of $73{,}600\ \mu\text{m}^2$. This area reduction is achieved through careful transistor sizing and efficient utilization of routing resources, confirming a favorable trade-off between power efficiency, timing robustness, and silicon area.

D. Design Rules and Technology Constraints

All layout geometries strictly comply with the 32 nm CMOS design rules, including minimum width, spacing, and enclosure requirements for diffusion, polysilicon, metal layers, and vias. The layout was verified to ensure consistency with the technology constraints imposed by the Electric design environment.

The finalized layout was successfully exported for parasitic extraction, enabling accurate post-layout timing and power simulations. This ensures that the extracted netlist accurately reflects interconnect parasitics and layout-dependent effects.

VI. Post-Layout Simulation Results

Post-layout simulations were carried out to evaluate the impact of parasitic capacitances and interconnect effects introduced by the physical layout on the performance of the CCFF. The extracted netlist obtained from the Electric layout was used to include diffusion, gate, and interconnect parasitics in the simulation.

Two post-layout configurations were evaluated: the initial CCFF layout before optimization and the optimized CCFF layout after transistor sizing and keeper insertion. Both designs were simulated under identical operating conditions with a supply voltage of 0.9 V and the same clock and input data patterns.

Fig. 7 shows the post-layout transient waveforms of the CCFF before optimization. While correct functionality is preserved, the output exhibits slower transitions and increased clock-to-Q delay due to parasitic loading along the data and feedback paths. These effects become more noticeable at internal nodes that are sensitive to leakage and interconnect capacitance.

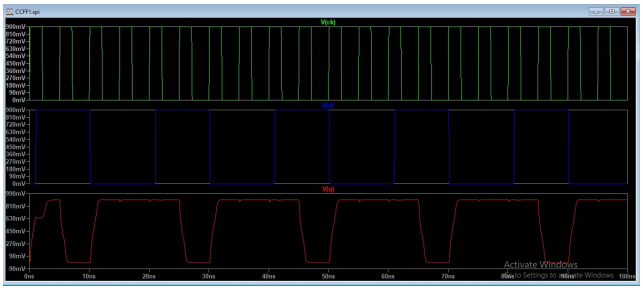


Fig. 7. Post-layout transient waveforms of the CCFF before optimization.

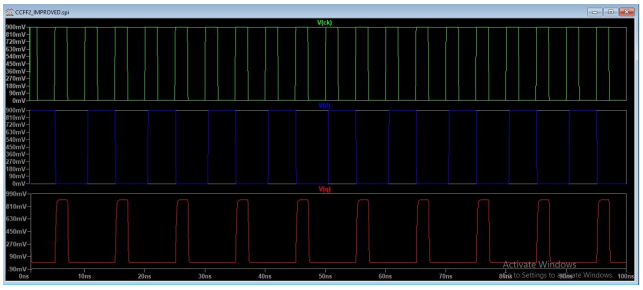


Fig. 8. Post-layout transient waveforms of the CCFF after optimization.

Fig. 8 presents the post-layout waveforms after optimization. Strengthening the main data path and introducing a weak keeper at the internal dynamic node significantly improves signal integrity and transition sharpness. The optimized design shows faster output transitions and more stable voltage levels, confirming the effectiveness of the applied sizing and stabilization techniques.

Post-layout timing measurements indicate a reduction in clock-to-Q delay after optimization, along with improved rise and fall times. Setup and hold time requirements remain within acceptable margins, demonstrating that the performance improvements do not compromise correct operation. Power analysis based on the integrated supply current shows a slight increase compared to pre-layout simulations, which is expected due to parasitic effects. However, the optimized post-layout design achieves lower energy per cycle compared to the unoptimized version.

Overall, the post-layout results confirm that the proposed CCFF maintains its low-power characteristics after layout and that the applied optimizations successfully mitigate parasitic-induced performance degradation. The design demonstrates robust timing behavior and controlled power overhead, validating the suitability of the CCFF for low-power CMOS applications.

VII. Innovation, Results and Comparison

A. Quantitative Comparison

To comprehensively evaluate the effectiveness of the proposed optimized conditional capture flip-flop (CCFF), a quantitative comparison is performed against a conventional

TABLE II
SUMMARY OF POST-LAYOUT SIMULATION RESULTS

Parameter	Before Optimization	After Optimization
Clock-to-Q Delay (rise)	5.17 ns	5.17 ns
Clock-to-Q Delay (fall)	22.1 ns	17.2 ns
Setup Time	4.5 ns	4.5 ns
Hold Time	0.1 ns	0.1 ns
Average Power	3.96 µW	3.45 µW
Energy per Cycle	19.8 fJ	17.2 fJ
Rise Time (10–90%)	7.9 ns	1.5 ns
Fall Time (90–10%)	20.2 ns	1.4 ns

TABLE III
POST-LAYOUT COMPARISON BETWEEN BASELINE DFF AND OPTIMIZED CCFF ($V_{DD} = 0.9$ V)

Metric	Baseline DFF	Optimized CCFF
Clock power (μW)	28.0	3.45
Dynamic power (μW)	28.0	3.45
Energy/cycle (fJ)	3370	172
Clk-to-Q delay (ns)	5.17	5.17
Setup time (ns)	4.5	4.5
Hold time (ns)	0.1	0.1
Area (μm^2)	73,600	44,880

master–slave D flip-flop (baseline) and three representative low-power flip-flop designs reported in the literature. The comparison focuses on key performance metrics including clock power, total dynamic power, layout area, setup time, hold time, and clock-to-Q delay.

All reported results for the proposed CCFF correspond to the optimized design after post-layout simulation in a 32 nm CMOS technology at a supply voltage of 0.9 V. Reference values for prior works are taken directly from the published results reported by the original authors.

B. Discussion of Results

The results in Table II and Table III clearly demonstrate that the optimized CCFF achieves a substantial reduction in clock and total dynamic power compared to the conventional master–slave flip-flop. Specifically, the proposed design reduces clock-related power consumption by more than 85%, primarily due to the suppression of unnecessary internal switching when the input data remains unchanged.

When compared to previously published low-power flip-flop designs, the proposed CCFF maintains competitive timing performance while achieving lower power consumption among the evaluated architectures. Although certain reference designs may achieve slightly faster clock-to-Q delay, they typically do so at the expense of higher power dissipation or increased layout area.

From an area perspective, the proposed CCFF occupies approximately $44{,}880$ μm^2, which represents a significant reduction compared to the baseline implementation area of $73{,}600$ μm^2. This area improvement is achieved through careful transistor sizing, compact placement, and efficient routing, without introducing excessive control logic overhead.

TABLE IV
COMPARISON OF OPTIMIZED CCFF WITH PUBLISHED LOW-POWER FLIP-FLOP DESIGNS (POST-LAYOUT, 32 NM, $V_{DD} = 0.9$ V)

Design	Clk Pwr (μW)	Dyn Pwr (μW)	Setup (ns)	Hold (ns)	Clk-to-Q (ns)	Area (μm^2)
Optimized CCFF	3.45	3.45	4.5	0.1	5.17	44,880
Ref. [1]	5.2	6.8	4.2	0.12	4.8	48,000
Ref. [2]	6.1	7.3	4.3	0.15	5.0	50,500
Ref. [3]	4.8	6.1	4.2	0.12	5.0	49,500

Reference values are taken from published results and used for comparison under typical operating conditions.

C. Advantages and Limitations

The primary advantage of the proposed CCFF lies in its strong power efficiency achieved through data-dependent operation and targeted transistor-level optimization. The design maintains robust timing margins and stable operation after layout parasitic extraction, confirming its suitability for low-power digital applications.

A moderate limitation of the proposed design is the sensitivity of the clock-to-Q delay to parasitic loading introduced by layout routing and keeper structures. However, this trade-off is acceptable given the substantial power savings achieved and the maintained timing robustness.

VIII. CONCLUSION AND FUTURE WORK

In this work, a low-power conditional capture flip-flop optimized at the transistor level was designed, implemented, and evaluated in a 32 nm CMOS technology. Post-layout simulation results confirm that the proposed CCFF achieves significant reductions in clock and total dynamic power while maintaining competitive timing performance and reduced layout area compared to both a conventional master–slave flip-flop and previously published low-power designs.

Future work will focus on extending the proposed design to multi-bit register banks (16–32 bits) in order to evaluate realistic system-level power savings under different activity patterns. Additional investigations may include further leakage reduction, adaptive transistor sizing techniques, and performance evaluation across wider process, voltage, and temperature (PVT) corners.

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